

VADODARA INSTITUTE OF COMPUTER APPLICATION
BCA –II, Semester–IV, Mid Semester Examination –summer 2025

Subject: Introduction to software engineering**Subject code: MINOR-3****Time: 1.30pm to 3.30pm****Instructions: All questions are compulsory.****Discipline: BCA****Date: 10/03/2026****Total Marks: 50**

Q-1 Answer the following Questions 10

- (a) Define Software Engineering. Explain the need for software engineering methods, tools, and procedures.
- (b) Describe the layers of the Software Engineering process with suitable examples.

OR

- (a) What is SEI-CMM? Explain its levels and significance in software process improvement.
- (b) Compare the Waterfall, Spiral, and Iterative development lifecycle models.

Q.2 Answer the following Questions 10

- (a) What is risk management in software projects? Explain various testing strategies such as top-down and bottom-up testing.
- (b) Explain modular design. Discuss coupling and cohesion with their types, merits, and demerits.

OR

- (a) Explain the steps, methods, and tools used in software prototyping.
- (b) What are PERT and CPM? Compare their use in project scheduling.

Q.3 Answer the following Questions 10

- (a) Write a short note on automated software estimation tools.
- (b) Explain the COCOMO model. Differentiate between basic, intermediate, and detailed COCOMO.

OR

- (a) Describe LOC and FP-based decomposition techniques for cost estimation.
- (b) Explain size-oriented and function-oriented metrics with examples.

Q.4 Answer the following Questions 10

- (a) Define software productivity metrics. Differentiate between direct and indirect productivity metrics.
- (b) What is Software Project Planning? Explain its scope, objectives, and required resources.

OR

- (a) **Write short notes on:**
 - a) RAD Model
 - b) Component-based Development Model
- (b) Explain the Prototyping Model. What are its advantages and disadvantages?

Q.5 Do as Direct 10

- 1. Which of the following is *not* a software development lifecycle model?
 - a) Waterfall Model
 - b) Spiral Model
 - c) Prototyping Model
 - d) Boolean Model

- 2 SEI-CMM primarily focuses on:
- a) Hardware design
 - b) Software process maturity
 - c) User interface standards
 - d) Risk elimination
- 3 Which metric is *function-oriented*?
- a) Lines of Code (LOC)
 - b) Function Points (FP)
 - c) KLOC
 - d) Cyclomatic Complexity
- 4 The COCOMO model is mainly used for:
- a) Software testing
 - b) Software cost estimation
 - c) Risk analysis
 - d) Debugging
- 5 In project scheduling, PERT stands for:
- a) Project Evaluation and Review Technique
 - b) Program Efficiency Review Test
 - c) Project Estimation and Resource Tracking
 - d) Performance Evaluation of Risk Techniques
- 6 Which coupling type is considered the *best* or most desirable?
- a) Content Coupling
 - b) Common Coupling
 - c) Control Coupling
 - d) Data Coupling
- 7 The prototyping model is most useful when:
- a) Requirements are well-defined
 - b) Requirements are unclear or evolving
 - c) The project is very large
 - d) Testing is not required
- 8 Cyclomatic complexity measures:
- a) Data structure efficiency
 - b) Number of modules in a system
 - c) Independent paths in a program
 - d) Execution time

Seat No.: _____

Enrolment No. _____

9 Top-down integration testing begins with:

- a)** Outer modules
- b)** Lowest-level modules
- c)** Middle modules
- d)** Main control module

10 Which of the following is an automated cost estimation tool?

- a)** MATLAB
- b)** Oracle
- c)** COCOMO II
- d)** GitHub

***** All the Best *****